SAFETY DATA SHEET

Section 1 - Identification

Product Name | **Yeast Nitrogen Base**

Product Code	**R458176**
Address	ThermoFisher Scientific Australia Pty Ltd 5 Caribbean Drive, Scoresby VICTORIA 3179, Australia
Emergency Tel.	**CHEMTREC®** **03 9757 4559 or +613 9757 4559**
Telephone / Fax Numbers	Tel: 1300 735 292 Fax: 1800 067 639
E-mail address	ANZinfo@thermofisher.com

Recommended Use Laboratory chemicals.

Uses advised against This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

Section 2 - Hazard(s) Identification

Classification under Safe Work Australia

Classified as not hazardous according to criteria of Safe Work Australia.

Physical hazards
No hazards identified

Health hazards
No hazards identified

Environmental hazards
No hazards identified

Label Elements

Other information

Toxic to terrestrial vertebrates
This product does not contain any known or suspected endocrine disruptors

Section 3 - Composition and Information on Ingredients

Component	CAS No	Weight %
Ammonium sulfate	7783-20-2	96.97
Histidine, L-, monohydrochloride monohydrate	5934-29-2	0.19
Iron(III) chloride	7705-08-0	Trace
Riboflavin	83-88-5	Trace
Sodium molybdate dihydrate	10102-40-6	Trace
DL-Methionine	59-51-8	0.39
DL-Tryptophan	54-12-6	0.39
Sodium chloride	7647-14-5	1.92
L-Glutamic acid, N-[4-[[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl]-	59-30-3	Trace
myo-Inositol	87-89-8	Trace
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-.beta.-alanine, calcium salt (2:1), (R)-	137-08-6	Trace
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride	58-56-0	Trace
Boric acid (H3BO3)	10043-35-3	Trace
Benzoic acid, 4-amino-, monosodium salt	555-06-6	Trace
(+)-Biotin	58-85-5	Trace
Copper (II) sulfate pentahydrate (1:1:5)	7758-99-8	Trace
Potassium iodide	7681-11-0	Trace
Zinc sulfate heptahydrate	7446-20-0	Trace
Manganese sulfate monohydrate	10034-96-5	Trace
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	67-03-8	Trace
3-Pyridinecarboxylic acid	59-67-6	Trace

Section 4 - First Aid Measures

Inhalation Remove to fresh air.

Ingestion Clean mouth with water and drink afterwards plenty of water.

Skin Contact Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.

Eye Contact Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.

Self-Protection of the First Aider No special precautions required.

First Aid Facilities Eyewash, safety shower and washroom.

Most important symptoms and effects No information available.

Notes to Physician Treat symptomatically.

Section 5 - Fire Fighting Measures

Suitable Extinguishing Media
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Extinguishing media which must not be used for safety reasons
No information available.

Specific Hazards Arising from the Chemical
Thermal decomposition can lead to release of irritating gases and vapors.

Special protective equipment and precautions for fire fighters
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

Section 6 - Accidental Release Measures

Emergency procedures
Ensure adequate ventilation.

Environmental Precautions
See Section 12 for additional Ecological Information. Do not flush into surface water or sanitary sewer system.

Methods for Containment and Clean Up

Clean-up methods - small spillage
Sweep up and shovel into suitable containers for disposal.

Clean-up methods - large spillage
Typically only supplied is small quantiites as packaged goods.
If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

Reference to Other Sections
Refer to protective measures listed in Sections 8 and 13.

Section 7 - Handling and Storage

Precautions for Safe Handling
Ensure adequate ventilation.

Conditions for Safe Storage, Including any Incompatibilities
Keep container tightly closed in a dry and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

Section 8 - Exposure Controls and Personal Protection

Exposure limits
AUS - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]
Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia
ACGIH - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.
UK - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.
DE - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

Component	Australia	New Zealand WEL	ACGIH TLV	The United Kingdom	Germany
Iron(III) chloride	TWA: 1 mg/m^3		TWA: 1 mg/m^3	STEL: 2 mg/m^3 15 min TWA: 1 mg/m^3 8 hr	
Sodium molybdate dihydrate	TWA: 5 mg/m^3		TWA: 0.5 mg/m^3	STEL: 10 mg/m^3 15 min TWA: 5 mg/m^3 8 hr	
Boric acid (H3BO3)			TWA: 2 mg/m^3		TWA: 0.5 mg/m^3 (8

			STEL: 6 mg/m³		Stunden). AGW - exposure factor 2 TWA: 10 mg/m³ (8 Stunden). MAK when boric acid and tetraborates are present together, the MAK value is 0.75 mg boron/m³ Höhepunkt: 10 mg/m³
Copper (II) sulfate pentahydrate (1:1:5)			TWA: 1 mg/m³	STEL: 2 mg/m³ 15 min TWA: 1 mg/m³ 8 hr	TWA: 0.01 mg/m³ (8 Stunden). MAK Höhepunkt: 0.02 mg/m³
Potassium iodide			TWA: 0.01 ppm		
Zinc sulfate heptahydrate					TWA: 0.1 mg/m³ (8 Stunden). MAK TWA: 2 mg/m³ (8 Stunden). MAK Höhepunkt: 0.4 mg/m³ Höhepunkt: 4 mg/m³
Manganese sulfate monohydrate	TWA: 1 mg/m³		TWA: 0.02 mg/m³ TWA: 0.1 mg/m³	STEL: 0.6 mg/m³ 15 min STEL: 0.15 mg/m³ 15 min TWA: 0.2 mg/m³ 8 hr TWA: 0.05 mg/m³ 8 hr	TWA: 0.2 mg/m³ (8 Stunden). AGW - exposure factor 8 TWA: 0.02 mg/m³ (8 Stunden). AGW - exposure factor 8 TWA: 0.2 mg/m³ (8 Stunden). MAK TWA: 0.02 mg/m³ (8 Stunden). MAK Höhepunkt: 1.6 mg/m³ Höhepunkt: 0.16 mg/m³

Biological limit values
This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Exposure Controls
Engineering Measures
None under normal use conditions.

Personal protective equipment

Eye Protection — Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

Hand Protection — Protective gloves

Glove material	Breakthrough time	Glove thickness	AUS/NZ Standard	Glove comments
Disposable gloves	See manufacturers recommendations	-	AS/NZS 2161	(minimum requirement)

Inspect gloves before use.
Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)
Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.
Remove gloves with care avoiding skin contamination.

Skin and body protection — Long sleeved clothing

Repiratory Protection — Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of repiratory protective devices

Recommended Filter type: — Particle filter (or AUS/NZ equivalent)

Hygiene Measures	Handle in accordance with good industrial hygiene and safety practice.
Environmental exposure controls	Prevent product from entering drains. Do not allow material to contaminate ground water system.

Section 9 - Physical and Chemical Properties

Information on basic physical and chemical properties

Appearance		
Physical State	Solid	
Odor	No information available	
Odor Threshold	No data available	
pH	No information available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flash Point	No information available	**Method -** No information available
Evaporation Rate	Not applicable	Solid
Flammability (solid,gas)	No information available	
Explosion Limits	No data available	
Vapor Pressure	No data available	
Vapor Density	Not applicable	Solid
Specific Gravity / Density	No data available	
Bulk Density	No data available	
Water Solubility	No information available	
Solubility in other solvents	No information available	

Partition Coefficient (n-octanol/water)

Component	**log Pow**
Ammonium sulfate	-5.1
Iron(III) chloride	-4
Riboflavin	-1.46
L-Glutamic acid, N-[4-[[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl]-	-2.41
myo-Inositol	-2.08
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-.beta.-alanine, calcium salt (2:1), (R)-	-5.35
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride	-0.7
Boric acid (H3BO3)	-0.757
Potassium iodide	0.04
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl-chloride, monohydrochloride	-3.04
3-Pyridinecarboxylic acid	2.34

Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
Viscosity	Not applicable	Solid
Explosive Properties	No information available	
Oxidizing Properties	No information available	

Other information

Section 10 - Stability and Reactivity

Reactivity None known, based on information available

Stability Stable under normal conditions.

Conditions to Avoid Heat, flames and sparks.

Incompatible Materials None known.

Hazardous Decomposition Products None under normal use conditions.

Hazardous Polymerization No information available.

Section 11 - Toxicological Information

Information on Toxicological Effects

Product Information

(a) acute toxicity;

Oral Based on available data, the classification criteria are not met
Dermal Based on available data, the classification criteria are not met
Inhalation Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Ammonium sulfate	2840 mg/kg (Rat)	LD50 > 2000 mg/kg (Rat)	
Iron(III) chloride	450 mg/kg (Rat) 316 mg/kg (Rat)		
Riboflavin	LD50 > 10 g/kg (Rat)		>5.4 mg/L/4h (rat)
Sodium chloride	LD50 = 3 g/kg (Rat)	LD50 > 10000 mg/kg (Rabbit)	LC50 > 42 mg/L (Rat) 1 h
L-Glutamic acid, N-[4-[[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl]-	10 g/kg (Mouse)		
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-.beta.-alanine, calcium salt (2:1), (R)-	LD50 > 10 g/kg (Rat)		
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride	4 g/kg (Rat)		
Boric acid (H3BO3)	2660 mg/kg (Rat)	> 2000 mg/kg (Rabbit)	Not listed
Copper (II) sulfate pentahydrate (1:1:5)	LD50 = 960 mg/kg (Rat)	LD50 > 8 g/kg (Rabbit)	
Potassium iodide	2779 mg/kg (Rat)	LD50 > 2000 mg/kg (Rat)	
Zinc sulfate heptahydrate	1260 mg/kg (Rat)		
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	LD50 = 3710 mg/kg (Rat)		
3-Pyridinecarboxylic acid	LD50 = 7 g/kg (Rat)	LD50 > 2000 mg/kg (Rat)	LC50 > 3.8 mg/L (Rat) 4 h

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable
Solid

Symptoms / effects,both acute and delayed No information available

Section 12 - Ecological Information

Ecotoxicity effects Contains a substance which is:. Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Ammonium sulfate	Cyprinus carpio: LC50: >460 mg/L/96h Brachydanio rerio: LC50: 420 mg/L/96h	EC50: 423 mg/L/24h LC50: 14 mg/L/48h	-	-
Iron(III) chloride	LC50: 20.95 - 22.56 mg/L, 96h semi-static (Pimephales promelas) LC50: = 20.26 mg/L, 96h semi-static (Lepomis macrochirus)	EC50: = 9.6 mg/L, 48h Static (Daphnia magna) EC50: = 27.9 mg/L, 48h (Daphnia magna)		
Riboflavin	Brachydanio rerio: LC50 >10000 mg/L/96h	EC50 >7.4 mg/L/48h		
Sodium chloride	Pimephals prome: LC50: 7650 mg/L/96h	EC50: 1000 mg/L/48h		
Boric acid (H3BO3)	Gambusia affinis: LC50: 5600 mg/L/96h	EC50: 115 - 153 mg/L, 48h (Daphnia magna)	-	-
Copper (II) sulfate pentahydrate (1:1:5)	Onchorhynchus mykiss: LC50 = 0.1-2.5 mg/L/96h	EC50 = 0.24 mg/L/48h		Photobacterium phosphoreum: EC50 = 0.25 mg/L/30min as Cu++ Photobacterium phosphoreum EC50= 1.3 mg/L/5 min as Cu++
Potassium iodide	Onchorhynchus mykiss: LC50: 3200 mg/L/120h	-	-	-
Zinc sulfate heptahydrate	1.9 mg/L LC50 96 h			
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	LC50 >100 mg/L/96h	EC50 >100 mg/L/48h		
3-Pyridinecarboxylic acid	LC50: = 520 mg/L, 96h	EC50: = 77 mg/L, 48h	EC50: = 89.93 mg/L,	= 160 mg/L EC50

	(Salmo trutta)	(Daphnia magna)	72h (Desmodesmus subspicatus)	Salmonella typhimurium 72 h = 2792.91 mg/L EC50 Tetrahymena pyriformis 60 h

Persistence and Degradability No information available

Persistence Persistence is unlikely.

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

Bioaccumulative Potential Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Ammonium sulfate	-5.1	No data available
Iron(III) chloride	-4	2756 - 9622 dimensionless
Riboflavin	-1.46	No data available
L-Glutamic acid, N-[4-[[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl]-	-2.41	No data available
myo-Inositol	-2.08	No data available
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-.beta.-alanine, calcium salt (2:1), (R)-	-5.35	No data available
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride	-0.7	No data available
Boric acid (H3BO3)	-0.757	0 dimensionless
Potassium iodide	0.04	No data available
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	-3.04	No data available
3-Pyridinecarboxylic acid	2.34	No data available

Mobility No information available.

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

Section 13 - Disposal Considerations

Waste from Residues/Unused Products Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

Contaminated Packaging Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

Other Information Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer.

Section 14 - Transport Information

IMDG/IMO Not regulated

ADG Not regulated

Component	Hazchem Code
Iron(III) chloride 7705-08-0 (Trace)	2X

IATA Not regulated

Environmental hazards No hazards identified

Special Precautions No special precautions required

Additional information None known

Section 15 - Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

National Regulations **Australia**

See section 8 for national exposure control parameters.

Standard for the Uniform Scheduling of Medicines and Poisons
Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

Component	Standard for the Uniform Scheduling of Medicines and Poisons
Iron(III) chloride - 7705-08-0	Schedule 2 listed Schedule 4 listed - in injectable preparations for human use Schedule 5 listed - for the treatment of animals except up to 1% of Iron oxides when present as an excipient;in preparations for injection except in preparations containing <=0.1% of Iron Schedule 5 listed - for the treatment of animals except up to 1% of Iron oxides when present as an excipient;in other preparations except in liquid or gel preparations containing <=0.1% of Iron, or in animal feeds or feed premixes Schedule 5 listed - for use as agricultural chemicals except in preparations containing <=4% of Iron Schedule 6 listed - except up to 1% of Iron oxides when present as an excipient. For the treatment of animals except: when included in Schedule 5, in liquid or gel preparations containing <=0.1% of Iron, or in animal feeds or feed premixes
DL-Tryptophan - 54-12-6	Schedule 4 listed - for human therapeutic use except in preparations labelled with a recommended daily dose of <=100 mg of Tryptophan
L-Glutamic acid, N-[4-[[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl]- - 59-30-3	Schedule 2 listed Schedule 4 listed - in preparations for human use for injection
Boric acid (H3BO3) - 10043-35-3	Schedule 4 listed - for human therapeutic use except: in preparations for internal use containing <=6 mg Boron per recommended daily dose, in preparations for dermal use containing <=0.35% of Boron, which are not for paediatric or antifungal use, or when present as an excipient Schedule 5 listed - except: a) when included in Schedule 4, or b) in cosmetic hand cleaning preparations when labelled with a warning to the following effect: NOT TO BE USED FOR CHILDREN UNDER 3 YEARS OF AGE, and if the concentration of free soluble Borates >1.5% (as Boric acid), with the words: NOT TO BE USED ON PEELING OR IRRITATED SKIN, or c) in cosmetic Talc preparations containing <=5% calculated as Boric acid when labelled with a warning to the following effect: NOT TO BE USED FOR CHILDREN UNDER 3 YEARS OF AGE, and if the concentration of free soluble Borates >1.5% (as Boric acid), with the words: NOT TO BE USED ON PEELING OR IRRITATED SKIN, or d) in cosmetic oral hygiene preparations containing <=0.1% calculated as Boric acid when labelled with a warning to the following effect: NOT TO BE SWALLOWED. NOT TO BE USED FOR CHILDREN UNDER 3 YEARS OF AGE, or e) in other cosmetic preparations containing <=3% calculated as Boric acid when labelled with a warning to the following effect: NOT TO BE USED FOR CHILDREN UNDER 3 YEARS OF AGE, and if the concentration of free soluble Borates >1.5% (as Boric acid), with the words: NOT TO BE USED ON PEELING OR IRRITATED SKIN, or f) in preparations, other than insect baits, containing <=6%, calculated as Boric acid
Copper (II) sulfate pentahydrate (1:1:5) - 7758-99-8	Schedule 4 listed - for human use except: when separately specified in these Schedules, in preparations for human internal use containing <=5 mg of Copper per recommended daily dose, or in other preparations containing <=5% of Copper compounds Schedule 5 listed - in animal feed additives except in preparations containing <=1% of Copper Schedule 6 listed - except: when separately specified in these Schedules, in preparations for human internal use containing <=5 mg of Copper per recommended daily dose, pigments where the solubility of the Copper compounds in water is <=1 g/L, in feed additives containing <=1% of Copper, or in other preparations containing <=5% of Copper compounds Schedule 6 listed - except when separately specified in these Schedules;in preparations for human internal use containing <=5 mg of Copper per recommended daily dose;pigments where the solubility of the Copper compounds in water is <=1 g/L;in feed additives containing <=1% of Copper, or in other preparations containing <=5% of Copper compounds
Zinc sulfate heptahydrate - 7446-20-0	Schedule 4 listed - for human internal use except in preparations with a recommended daily dose of

	<=25 mg of Zinc, or in preparations with a recommended daily dose of between 25-50 mg of Zinc when compliant with the requirements of the Required Advisory Statements for Medicine Labels
3-Pyridinecarboxylic acid - 59-67-6	Schedule 3 listed Schedule 4 listed - for human therapeutic use except: when contained in other schedules, in preparations containing <=100 mg of Nicotinic acid per dosage unit, or Nicotinamide

Australian Industrial Chemicals Introduction Scheme (AICIS)

Component	Australian Industrial Chemicals Introduction Scheme (AICIS)	Additional information
Ammonium sulfate - 7783-20-2	Present	-
Histidine, L-, monohydrochloride monohydrate - 5934-29-2	Present	-
Iron(III) chloride - 7705-08-0	Present	-
Riboflavin - 83-88-5	Present	-
Sodium molybdate dihydrate - 10102-40-6	Present	-
DL-Methionine - 59-51-8	Present	-
DL-Tryptophan - 54-12-6	Present	-
Sodium chloride - 7647-14-5	Present	-
L-Glutamic acid, N-[4-[[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl]- - 59-30-3	Present	-
myo-Inositol - 87-89-8	Present	-
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-.beta.-alanine, calcium salt (2:1), (R)- - 137-08-6	Present	-
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride - 58-56-0	Present	-
Boric acid (H3BO3) - 10043-35-3	Present	-
Benzoic acid, 4-amino-, monosodium salt - 555-06-6	Present	-
(+)-Biotin - 58-85-5	Present	-
Copper (II) sulfate pentahydrate (1:1:5) - 7758-99-8	Present	-
Potassium iodide - 7681-11-0	Present	-
Zinc sulfate heptahydrate - 7446-20-0	Present	-
Manganese sulfate monohydrate - 10034-96-5	Present	-
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride - 67-03-8	Present	-
3-Pyridinecarboxylic acid - 59-67-6	Present	-

Australian - Illicit Drug Precursors/Reagents Substance List
This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

Chemicals of Security Concern
This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

National pollutant inventory Not applicable

Prohibition or notification/licensing requirements
Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

International Inventories

Component	AICS	NZIoC	EINECS	ELINCS	TSCA	DSL	NDSL	PICCS	ENCS	ISHL	IECSC	KECL
Ammonium sulfate	X	X	231-984-1	-	X	X	-	X	X	X	X	KE-01743
Histidine, L-, monohydrochloride monohydrate	X	X	-	-	-	-	-	X	-		X	-
Iron(III) chloride	X	X	231-729-4	-	X	X	-	X	X	X	X	KE-21134
Riboflavin	X	X	201-507-1	-	X	X	-	X	X	X	X	KE-11845
Sodium molybdate dihydrate	X	X	-	-	-	-	-	X	X	X	X	-
DL-Methionine	X	X	200-432-1	-	X	X	-	X	X	X	X	KE-14-0064
DL-Tryptophan	X	X	200-194-9	-	X	X	-	X	X	X	-	2014-3-5933
Sodium chloride	X	X	231-598-3	-	X	X	-	X	X	X	X	KE-31387
L-Glutamic acid, N-[4-[[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl]-	X	X	200-419-0	-	X	X	-	X	X		X	KE-01305
myo-Inositol	X	X	201-781-2	-	X	X	-	X	X	X	X	KE-21013
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-.beta.-alanine, calcium salt (2:1), (R)-	X	X	205-278-9	-	X	X	-	X	X	X	X	KE-10821
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride	X	X	200-386-2	-	X	X	-	X	X	X	X	KE-20695
Boric acid (H3BO3)	X	X	233-139-2	-	X	X	-	X	X	X	X	KE-03499
Benzoic acid, 4-amino-, monosodium salt	X	-	209-080-3	-	X	-	X	-	X	X	X	-
(+)-Biotin	X	X	200-399-3	-	X	X	-	X	X	X	X	KE-18590
Copper (II) sulfate pentahydrate (1:1:5)	X	X	-	-	-	-	-	X	X		X	-
Potassium iodide	X	X	231-659-4	-	X	X	-	X	X	X	X	KE-29149
Zinc sulfate heptahydrate	X	X	-	-	-	X	-	X	X		X	-
Manganese sulfate monohydrate	X	X	-	-	-	-	-	X	X	X	X	-
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	X	X	200-641-8	-	X	X	-	X	X	X	X	KE-01482
3-Pyridinecarboxylic acid	X	X	200-441-0	-	X	X	-	X	X	X	X	KE-29937

Legend: X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

International Regulations

Ozone Depletion Potential This product does not contain any known or suspected substance

Persistent Organic Pollutant This product does not contain any known or suspected substance

Rotterdam Convention (PIC) Not applicable

Basel convention on the control of transboundary movements of hazardous wastes and their dispoal
Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

Component	Basel Convention (Hazardous Waste)	Australian Hazardous Waste Act - Categories of Wastes to Be Controlled
Copper (II) sulfate pentahydrate (1:1:5) - 7758-99-8	Annex I - Y22	Y22
Zinc sulfate heptahydrate - 7446-20-0	Annex I - Y23	Y23

Component	CAS No	OECD HPV	Restriction of Hazardous Substances (RoHS)	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Ammonium sulfate	7783-20-2	Listed	Not applicable	Not applicable	Not applicable
Histidine, L-, monohydrochloride monohydrate	5934-29-2	Not applicable	Not applicable	Not applicable	Not applicable
Iron(III) chloride	7705-08-0	Listed	Not applicable	Not applicable	Not applicable
Riboflavin	83-88-5	Listed	Not applicable	Not applicable	Not applicable
Sodium molybdate dihydrate	10102-40-6	Not applicable	Not applicable	Not applicable	Not applicable
DL-Methionine	59-51-8	Listed	Not applicable	Not applicable	Not applicable
DL-Tryptophan	54-12-6	Not applicable	Not applicable	Not applicable	Not applicable
Sodium chloride	7647-14-5	Listed	Not applicable	Not applicable	Not applicable
L-Glutamic acid, N-[4-[[(2-amino-1,4-dihydro-4-oxo-6-pteridinyl)methyl]amino]benzoyl]-	59-30-3	Not applicable	Not applicable	Not applicable	Not applicable
myo-Inositol	87-89-8	Not applicable	Not applicable	Not applicable	Not applicable
N-(2,4-Dihydroxy-3,3-dimethyl-1-oxobutyl)-.beta.-alanine, calcium salt (2:1), (R)-	137-08-6	Not applicable	Not applicable	Not applicable	Not applicable
3,4-Pyridinedimethanol, 5-hydroxy-6-methyl-, hydrochloride	58-56-0	Listed	Not applicable	Not applicable	Not applicable
Boric acid (H3BO3)	10043-35-3	Listed	Not applicable	Not applicable	Not applicable
Benzoic acid, 4-amino-, monosodium salt	555-06-6	Not applicable	Not applicable	Not applicable	Not applicable
(+)-Biotin	58-85-5	Not applicable	Not applicable	Not applicable	Not applicable
Copper (II) sulfate pentahydrate (1:1:5)	7758-99-8	Listed	Not applicable	Not applicable	Not applicable
Potassium iodide	7681-11-0	Listed	Not applicable	Not applicable	Not applicable
Zinc sulfate heptahydrate	7446-20-0	Not applicable	Not applicable	Not applicable	Not applicable
Manganese sulfate monohydrate	10034-96-5	Not applicable	Not applicable	Not applicable	Not applicable
Thiazolium, 3-[(4-amino-2-methyl-5-pyrimidinyl)methyl]-5-(2-hydroxyethyl)-4-methyl- chloride, monohydrochloride	67-03-8	Not applicable	Not applicable	Not applicable	Not applicable
3-Pyridinecarboxylic acid	59-67-6	Listed	Not applicable	Not applicable	Not applicable

Authorisation/Restrictions according to EU REACH

Component	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Ammonium sulfate	-	Use restricted. See item 65. (see link for restriction details)	-
Riboflavin	-	Use restricted. See item 75. (see link for restriction details)	-
Boric acid (H3BO3)	-	Use restricted. See item 30. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	SVHC Candidate list - 233-139-2 - Toxic for reproduction, Article 57c
Copper (II) sulfate pentahydrate (1:1:5)	-	Use restricted. See item 75. (see link for restriction details)	-
Zinc sulfate heptahydrate	-	Use restricted. See item 75. (see link for restriction details)	-

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

https://echa.europa.eu/authorisation-list
https://echa.europa.eu/substances-restricted-under-reach
https://echa.europa.eu/candidate-list-table

Section 16 - Other Information

Legend

AICS - Australian Inventory of Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
IECSC - Chinese Inventory of Existing Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
NZS 5433:2020 - Transport of Dangerous Goods on Land
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
WEL - Workplace Exposure Limit
DNEL - Derived No Effect Level
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative
VOC - (Volatile Organic Compound)
NZIoC - New Zealand Inventory of Chemicals
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
ENCS - Japanese Existing and New Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
CAS - Chemical Abstracts Service
ACGIH - American Conference of Governmental Industrial Hygienists Predicted No Effect Concentration (PNEC)
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
ADG - Australian Code for the Transport of Dangerous Goods by Road and Rail
OECD - Organisation for Economic Co-operation and Development
LC50 - Lethal Concentration 50%
ATE - Acute Toxicity Estimate
RPE - Respiratory Protective Equipment
NOEC - No Observed Effect Concentration
BCF - Bioconcentration factor
PBT - Persistent, Bioaccumulative, Toxic

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Revision Date	05-Jul-2023
Revision Summary	Not applicable.

This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).

Disclaimer

End of Safety Data Sheet